Smart Traffic Lights That Learn

A Systematic Literature Review of Reinforcement Learning for Traffic Signal Control — From Simulation to Real Streets. Project documentation: Steps 1–27

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Project overview and status

This repository holds the SLR as a reproducible research project. It contains the planning and protocol documents, the locked search strategy, and a Python pipeline (`slr/`) for Steps 13–27. The pipeline produces the master evidence register, the PRISMA 2020 flow diagram, the quality-assessment and extraction sheets, and all synthesis tables and figures, computed directly from the database exports and the reviewers' decisions. The original project proposal is `docs/00_Project_Proposal.docx`.

Status of Steps 1–27

Step	Deliverable	Status
1 Topic	`docs/phase1_planning/01_Topic_Feasibility_Note.md`	✅ Done
2 Scoping search	`docs/phase1_planning/02_Scoping_Search_Report.md`	✅ Done (existing reviews, terminology, platforms, early gaps)
3 Scope	`docs/phase1_planning/03_Scope_Statement.md`	✅ Done
4–5 Objectives, RQs	`docs/phase1_planning/04_05_Objectives_and_Research_Questions.md`	✅ Done (6 RQs + Validation Realism Level scale)
6 Protocol	`docs/phase1_planning/06_Review_Protocol.md`	✅ v1.0 (register on OSF before searching)
7 Concept matrix	`docs/phase2_search/07_Search_Concept_Matrix.md`	✅ Done
8 Databases	`docs/phase2_search/08_Database_Selection.md`	✅ Scopus + WoS CC + IEEE Xplore
9 Search strings	`docs/phase2_search/09_Database_Search_Strings.md`, `data/protocol/search_strings.json`	✅ Written for all three databases
10 Pilot	`docs/phase2_search/10_Pilot_Search.md`, `data/protocol/known_relevant_papers.csv`, `python -m slr pilot`	🔶 Procedure, 20-paper gold set and recall tool ready. **The team runs the pilot.**
11 Lock	`docs/phase2_search/11_Search_Protocol_v1.0_LOCKED.md`	🔶 Sign after the pilot
12 Multi-wave plan	`docs/phase2_search/12_Multi_Wave_Search_Plan.md`	✅ Plan ready; waves are tracked automatically
13 Master register	`outputs/Master_Evidence_Register.xlsx` (15 sheets), `python -m slr workbook`	✅ Built (empty until the searches are imported)
14 Deduplication	`python -m slr dedup`	✅ Tool ready and tested
15 Title/abstract screening	`python -m slr screen-ta`	✅ Tool ready (calibration sample, κ, conservative rule)
16 Full-text screening	`python -m slr screen-ft`	✅ Tool ready (E1–E10 reasons, conflicts, stable paper IDs)
17 PRISMA diagram	`python -m slr prisma`	✅ Generated from the records (no manual counts)
18 Quality assessment	`docs/phase4_quality/18_Quality_Assessment.md`, `python -m slr qa`	✅ Rubric + tool (weighted κ)
19–21 Extraction	`docs/phase5_extraction/19-21_Data_Extraction.md`, `data/protocol/codebook.csv`, `python -m slr extract`	✅ Codebook, validation, traceable evidence register
22–27 Synthesis	`docs/phase6_synthesis/22-27_Synthesis_Plan.md`, `python -m slr synthesis`	✅ Taxonomy, method comparison, bibliometrics, statistics, meta-analysis rule, dependence control

What the tools cannot do for you

- **Running the database searches.** Scopus and Web of Science need the university subscription. Export the results as described in Document 09.
- **Screening, quality and extraction decisions.** These are human judgements by R1 and R2. The pipeline organises them, checks them, measures agreement and computes everything downstream.

No PRISMA count, quality score or extracted number in this repository is estimated or invented. Each one is computed from the files in `data/`, as PRISMA requires.

Quick start

```
        pip install -r requirements.txt
        python -m pytest -q             # tests on synthetic fixture exports

        # Step 10 — pilot
        #   put pilot exports in data/raw/<db>/pilot_*.csv|txt
        python -m slr pilot

        # Steps 13-27 — after the search is locked
        #   1. put exports in data/raw/scopus|wos|ieee|other/ and fill data/protocol/search_log.csv
        python -m slr ingest
        python -m slr dedup             # then confirm data/records/dedup_candidates.csv (is_duplicate Y/N)
and re-run
        python -m slr screen-ta         # reviewers fill data/screening/title_abstract.csv; re-run; --stats
for kappa
        python -m slr screen-ft         # reviewers fill data/screening/full_text.csv; re-run
        python -m slr qa                # reviewers fill data/quality/quality_assessment.csv; re-run
        python -m slr extract           # reviewers fill data/extraction/*.csv; re-run (validates +
computes RI)
        python -m slr prisma            # outputs/prisma/prisma_flow.png
        python -m slr synthesis         # outputs/synthesis/synthesis_report.md, tables/, figures/
        python -m slr workbook          # outputs/Master_Evidence_Register.xlsx

        python -m slr all               # every stage that has input, in order

        python scripts/build_report.py  # all documents in one file: outputs/SLR_Project_Documentation.docx
+ .pdf
```

Every command can be re-run at any time. Reviewer input is never overwritten.

Repository layout

```
docs/                   planning, protocol and procedure documents (Steps 1-27)
data/protocol/          locked search strings, search log, gold set, codebook, amendments, templates
data/raw/<source>/      unmodified database exports (input)
data/records/           raw records, deduplicated master records, dedup log/candidates (generated)
data/screening/         title/abstract + full-text sheets, included studies (reviewers fill decisions)
data/quality/           quality assessment sheet (reviewers fill scores)
data/extraction/        technical + quantitative extraction sheets (reviewers fill)
data/synthesis/         RQ evidence and research-gap tables (reviewers fill)
outputs/                PRISMA diagram, synthesis report, tables, figures, master workbook (generated)
slr/                    pipeline source code
tests/                  unit + end-to-end tests with synthetic fixtures (not review evidence)
```

Research questions

1. **RQ1 Landscape** — How has RL-based traffic signal control research evolved between 2019 and 2026?
2. **RQ2 Methods** — Which RL formulations are used (family, agents, state, action, reward)?
3. **RQ3 Validation** — How are the methods evaluated, and how realistic is the validation (VRL 0 synthetic simulation → VRL 5 sustained deployment)?
4. **RQ4 Performance** — How much do they improve waiting time, queue length and travel time over fixed-time, actuated, Max-Pressure and RL baselines, and does the improvement depend on validation realism?
5. **RQ5 Barriers** — What stops deployment on real streets, and what mitigations are proposed?
6. **RQ6 Gaps** — Which gaps remain, and what should future research focus on?

Part I — Review planning (Steps 1–6)

Document 01 — Topic Feasibility Note (Step 1)

Item	Value
Project	Smart Traffic Lights That Learn
Authors	Rakhimzhanov Amangeldy, Nurkin Aidar
Document version	1.0
Status	Final (planning phase)

1. Topic decomposition

Level	Content
Broad field	Intelligent transportation systems / smart systems / artificial intelligence
Technology / method	Reinforcement learning (RL), including deep RL (DRL) and multi-agent RL (MARL)
Application / context	Traffic signal control (TSC) at urban signalised intersections and networks of intersections
Specific angle	Gap between simulation results and real-world (field) deployment

2. Provisional title

"Smart Traffic Lights That Learn: A Systematic Literature Review of Reinforcement Learning for Traffic Signal Control — From Simulation to Real Streets"

Alternative (guideline template form): "Reinforcement Learning for Adaptive Traffic Signal Control: A Systematic Review of Methods, Validation Realism, Performance, Deployment Barriers and Future Research Directions"

3. Motivation

- Most urban signals still run fixed-time or simple actuated plans that do not adapt to real demand.
- RL lets a signal controller learn a timing policy from interaction (sense → decide → act → reward), which makes it a textbook *smart system* (perception, decision, actuation in a closed loop).
- Hundreds of papers report reductions in waiting time, queue length and travel time **in simulation**, but published field deployments remain very rare (Chen, Fang & Sadeh 2022; Müller & Sabatelli 2022; see Document 02).
- There is no recent systematic review that codes **how realistic the validation of each study is** and links it to reported gains and deployment barriers.

4. Broad research problem

Why do RL-based traffic signal controllers that perform well in simulation rarely reach real streets, and what evidence exists on methods, validation practice, performance and deployment barriers that could close this gap?

5. Feasibility questions (guideline Step 1)

#	Question	Assessment
1	Enough published research?	**Yes.** The topic is very active (KDD, CIKM, AAAI, NeurIPS, IEEE T-ITS, IEEE ITSC, Transportation Research Part C, Expert Systems with Applications…). An earlier SLR (Noaeen et al., 2022) already found 160 network-level articles up to March 2020 alone.
2	Scientifically important?	**Yes.** Congestion costs time, fuel, emissions and safety; signal control is a cheap lever because the infrastructure already exists.

#	Question	Assessment
3	Similar recent SLR published?	**Partly.** Noaeen et al. (2022, ESWA) — systematic, but corpus ends March 2020 and focus is network-level. Haydari & Yilmaz (2022, IEEE T-ITS) — narrative survey of DRL in ITS. Zhang, Cui & Ma (2024, Transport Reviews) — focus on connected/automated vehicles. Michailidis et al. (2025, Infrastructures) — review of high-impact methods. Chen, Fang & Sadeh (2022) — position paper on deployment barriers, not systematic.
4	Meaningful new contribution?	**Yes.** (a) covers 2019–2026, including the sim-to-real, LLM-based and safety-constrained work published after 2022; (b) introduces an explicit **Validation Realism Level (VRL 0–5)** coding of every study; (c) relates reported improvements to the baseline strength and validation realism; (d) controls double counting of shared benchmark datasets (Study Family IDs).
5	Can the topic be defined for systematic searching?	**Yes.** Two well-delimited concepts (RL × traffic signals) with stable vocabulary; see Document 07.
6	Quantitative / methodological comparison possible?	**Yes, descriptively.** Most papers report average waiting time, queue length, travel time or delay against fixed-time / actuated / Max-Pressure baselines, so relative improvement can be computed. A formal meta-analysis is unlikely to be appropriate (heterogeneous scenarios, variance rarely reported); see Document 26.
7	Important gaps likely?	**Yes.** Field validation, robustness to sensor failure, safety, pedestrians/cyclists/public transport, interpretability and regulatory compliance.

6. Preliminary keywords

reinforcement learning; deep reinforcement learning; multi-agent reinforcement learning; Q-learning; deep Q-network (DQN); actor–critic; policy gradient; PPO; traffic signal control; traffic light control; adaptive traffic signal control; signalised/signalized intersection; signal timing; signal phase; sim-to-real; field deployment; real-world deployment.

7. Preliminary databases

Core: **Scopus**, **Web of Science Core Collection**, **IEEE Xplore**. Supplementary (not counted as databases in PRISMA): Google Scholar and publisher pages for citation chaining, and ACM Digital Library as a cross-check for KDD/CIKM papers.

8. Review period

1 January 2019 – date of the final search (2026). Justification is given in Document 03 §3.5.

9. Expected contribution

1. An up-to-date, reproducible map of RL-for-TSC research 2019–2026.
2. A taxonomy of RL formulations (state, action, reward, agent architecture, algorithm family).
3. The first systematic coding of **validation realism** (simulation → calibrated simulation → hardware-in-the-loop → field test → sustained operation).
4. A descriptive quantitative synthesis of reported improvements, stratified by baseline type and validation realism, with dependence control.
5. A structured catalogue of deployment barriers and a prioritised research agenda.

Document 02 — Preliminary Scoping Search Report (Step 2)

> **Important distinction.** This scoping search was used only to *design* the review (terminology, scope, existing reviews, likely gaps). None of the records found here are counted in the PRISMA flow. The formal evidence comes only from the locked database searches (Documents 09–12).

Item	Value
Date	17 Oct 2026
Tool	General web search engine (exploratory, non-reproducible by design)
Queries run	8 (listed in §1)

1. Exploratory queries

1. systematic review reinforcement learning traffic signal control real-world deployment
2. reinforcement learning traffic signal control field deployment real intersection test 2023 2024
3. sim-to-real transfer reinforcement learning traffic signal control
4. survey deep reinforcement learning traffic signal control (Haydari & Yilmaz; Wei, Zheng, Gayah & Li)
5. benchmark traffic signal control reinforcement learning RESCO LibSignal CityFlow SUMO
6. multi-agent reinforcement learning traffic signal CoLight PressLight MPLight MA2C
7. “adaptive traffic signal control” reinforcement learning systematic review 2024 2025
8. field test / field trial deep reinforcement learning traffic signal controller real-world city
9. large language model traffic signal control LLMLight

2. Terminology and synonyms found

Concept	Variants observed
Reinforcement learning	reinforcement learning, deep reinforcement learning (DRL/deep RL), multi-agent RL (MARL), Q-learning, deep Q-network (DQN, DDQN, D3QN), actor–critic (A2C, A3C), policy gradient, PPO / IPPO / MAPPO, DDPG, SAC, meta-RL, hierarchical RL, offline RL
Traffic signal	traffic signal control (TSC), adaptive traffic signal control (ATSC), traffic light control, intelligent traffic light, signalised/signalized intersection, signal timing / timing plan, signal phase / phase selection / phase duration, cycle length, green split, arterial coordination, network-level control
Deployment	real-world deployment, field test/trial, sim-to-real / sim2real, transfer, grounded action transformation, domain randomisation, hardware-in-the-loop, digital twin, real intersection
Emerging	large language model (LLM) agents for TSC (e.g., LLMLight), safe RL / action masking, robustness to detector failure

3. Major existing reviews (to position our contribution)

Review	Type	Coverage	Relation to our SLR
Noaeen et al. (2022), *Reinforcement learning in urban network traffic signal control: A systematic literature review*, Expert Systems with Applications	SLR	160 articles, 1994 – Mar 2020, network-level	Closest prior SLR; we extend the period and add validation-realism coding
Haydari & Yilmaz (2022), *Deep Reinforcement Learning for Intelligent Transportation Systems: A Survey*, IEEE T-ITS 23(1)	Narrative survey	DRL in ITS incl. TSC	Background for taxonomy
Wei, Zheng, Gayah & Li (2019/2021), *A Survey on Traffic Signal Control Methods* (arXiv 1904.08117) / *Recent advances in RL for TSC* (SIGKDD Explorations)	Narrative survey	Classical and RL TSC	Background for method families
Zhang, Cui & Ma (2024), Transport Reviews 44(6)	Systematic review	RL-TSC with CAVs	Different focus (CAV)

Review	Type	Coverage	Relation to our SLR
Michailidis et al. (2025), *Traffic Signal Control via Reinforcement Learning: A Review on Applications and Innovations*, Infrastructures 10(5):114	Review	High-impact RL-TSC methods	Recent; not deployment-centred
Chen, Fang & Sadeh (2022), *The Real Deal: A Review of Challenges and Opportunities in Moving Reinforcement Learning-Based Traffic Signal Control Systems Towards Reality* (ATT'22)	Position review	Deployment barriers	Motivates our RQ5; not systematic
TU Delft repository review on deep Q-learning controllers	Systematic review	DQN design choices	Narrow algorithm focus

4. Dominant technologies, platforms and benchmarks

- **Simulators:** SUMO and CityFlow dominate; also VISSIM, Aimsun, CityFlow-based LibSignal, RESCO (SUMO-based benchmark: Ingolstadt, Cologne scenarios).
- **Benchmark datasets reused across many papers:** Jinan (3×4), Hangzhou (4×4), New York (16×3, 28×7), Shanghai, Monaco, Ingolstadt, Cologne. → strong dependence between publications (handled in Step 27).
- **Landmark method families:** single-intersection DQN; phase-competition (FRAP); pressure-based rewards (PressLight, MPLight, Advanced-XLight); graph/attention communication (CoLight, CoSLight); meta-RL (MetaLight); universal/attention models (AttendLight); MARL with actor–critic (MA2C); sim-to-real (UGAT, GAT-CASE23, PromptGAT); LLM agents (LLMLight).
- **Common baselines:** fixed-time (Webster), actuated, Max-Pressure, SOTL, SCOOT/SCATS-like adaptive systems (rare).

5. Typical study designs and quantitative variables

- Design: algorithm proposal + simulation experiment versus baselines; occasional real-data-calibrated simulation; very rare field tests.
- Variables: average travel time, average waiting time, average delay, queue length, throughput, number of stops, fuel/CO_2 emissions, safety surrogates (conflicts, TTC), training episodes / convergence.
- Dispersion (SD over random seeds) is inconsistently reported.

6. Potential research gaps already visible

1. Field validation (very few peer-reviewed live tests; scoping searches found mostly simulation work and position papers).
2. Robustness to detector noise/failure and partial observability.
3. Safety constraints, minimum green / pedestrian clearance compliance.
4. Multimodality (pedestrians, cyclists, public-transport priority, emergency vehicles).
5. Interpretability and acceptance by traffic engineers.
6. Comparison against strong adaptive baselines (Max-Pressure, SCATS/SCOOT) rather than fixed-time only.
7. Reproducibility (code, seeds, scenario files).

7. Sources consulted (scoping only)

- https://arxiv.org/abs/2206.11996 (Chen, Fang & Sadeh — The Real Deal)
- https://arxiv.org/abs/2103.16223 (Müller et al. — Towards Real-World Deployment of RL for TSC)
- https://arxiv.org/pdf/2206.10122 (Müller & Sabatelli — Safe and psychologically pleasant TSC with action masking)
- https://datalearner.com/academic/journal-papers/0957-4174/volumes-and-issues/403/paper-detail/18449 (Noaeen et al. SLR record)
- https://arxiv.org/abs/2005.00935 (Haydari & Yilmaz survey)
- https://arxiv.org/pdf/1904.08117 (Wei et al. survey)
- https://ideas.repec.org/a/taf/transr/v44y2024i6p1187-1208.html (Zhang, Cui & Ma 2024)
- https://doi.org/10.3390/infrastructures10050114 (Michailidis et al. 2025)
- https://trid.trb.org/View/2713949 (RL for ATSC: an overview)
- https://repository.tudelft.nl/file/File_dd04ad92-3d98-4c41-998e-45c1c3b6fe01 (TU Delft DQN review)
- https://arxiv.org/abs/2307.12388v2 (UGAT sim-to-real)

- https://ojs.aaai.org/index.php/AAAI/article/view/27758 (PromptGAT, AAAI 2024)
- https://digitalcommons.njit.edu/fac_pubs/2219 (Sim2Real transfer for TSC, CASE 2023)
- https://arxiv.org/pdf/2211.10649 (LibSignal)
- https://github.com/behradkoohy/RESCO (RESCO benchmark)
- https://arxiv.org/abs/1905.05717 (CoLight)
- https://arxiv.org/html/2405.17152v2 (CoSLight)
- https://arxiv.org/pdf/2204.12190 (UniComm)
- https://arxiv.org/abs/2312.16044v5 (LLMLight)
- https://arxiv.org/pdf/2008.11634 (Egea et al. — reward functions under real-world limitations)
- https://www.nature.com/articles/s41598-025-18449-1 (Hierarchical RL-based TSC, 2025)
- https://traffic-signal-control.github.io/ (community resource list)

> Bibliographic details of these items must be verified against the publisher record before being cited in the manuscript.

Document 03 — Scope Statement (Step 3)

Formal scope statement

> This systematic review investigates **peer-reviewed primary studies** concerning **reinforcement learning (RL)–based controllers** applied to **road traffic signals at one or more intersections**, for the purpose of **adaptively selecting or timing signal phases to improve traffic performance**, published between **1 January 2019 and the date of the final search (2026)**, with particular attention to **how realistically the controllers were validated (simulation → field)**.

3.1 Subject scope

The phenomenon studied is **learning-based adaptive signal control**: a controller that *learns* its policy (phase choice, phase duration, cycle length, green split or offset) through RL from a reward signal.

In scope:

- tabular RL (Q-learning, SARSA), deep value-based (DQN family), policy-gradient / actor–critic (A2C, A3C, PPO, DDPG, SAC…), multi-agent RL, hierarchical RL, meta-RL, offline RL, model-based RL;
- hybrid methods where RL is the decision layer (e.g., RL + graph neural network, RL + fuzzy rules, RL + Max-Pressure reward, LLM agents whose policy is refined with RL).

Out of scope:

- signal optimisation by non-RL methods only (genetic algorithms, MPC, fuzzy logic, supervised prediction) — unless RL is the evaluated controller;
- LLM/prompting-only controllers with no RL component;
- RL used only as a minor comparison baseline.

3.2 Application scope

In scope: urban and suburban **road intersections** (isolated, arterial corridor, grid/network), including signal control for mixed traffic, connected/automated vehicles, pedestrians, cyclists, public-transport or emergency-vehicle priority **when RL controls the signal**.

Out of scope: ramp metering only, variable speed limits only, lane-use control only, routing/navigation, autonomous-vehicle driving policies, railway/air/maritime traffic, telecommunications “traffic” and “signals”.

3.3 Technology scope

Any sensing source (loop detectors, cameras, radar, connected-vehicle data, simulation ground truth) and any evaluation environment (SUMO, CityFlow, VISSIM, Aimsun, Paramics, custom simulators, hardware-in-the-loop, field).

3.4 Geographic scope

No geographic restriction. Country of authors and location of studied networks are extracted as variables (RQ1).

3.5 Publication period

1 January 2019 – final search date (2026).

Justification:

1. **Continuity with the previous SLR.** Noaeen et al. (2022) cover up to March 2020; starting in 2019 gives a deliberate one-year overlap to check consistency, and our focus (validation realism, deployment) is different.
2. **Technological shift.** 2019 is when open, large-scale RL-TSC tooling became standard (CityFlow released 2019; real-data benchmark datasets of Jinan/Hangzhou/New York; landmark DRL models PressLight, CoLight, FRAP published 2019). This made studies comparable and sharply increased publication volume.
3. **Deployment debate.** Sim-to-real, robustness and deployment-barrier papers appear from 2021 onwards; a 2019 start captures the full arc.

3.6 Publication type

Type	Decision	Reason
Journal articles (incl. early access/articles in press)	**Include**	Peer-reviewed primary evidence
Full conference papers (proceedings, ≥ 4 pages)	**Include**	Key venues in this field are conferences (KDD, CIKM, AAAI, NeurIPS, ITSC)
Book chapters	Exclude	Rarely primary, inconsistent peer review
Reviews / surveys	Exclude from evidence base; **used for citation chaining (Wave 3) and Discussion**	Not primary studies
Preprints (arXiv etc.), theses, reports, patents, grey literature	Exclude	Not peer-reviewed; a preprint is replaced by its peer-reviewed version if one exists
Extended abstracts, posters, editorials	Exclude	Insufficient technical information

Language: **English only** (documented as a limitation).

Documents 04–05 — Review Objectives and Research Questions (Steps 4–5)

Step 4 — Main objective

> The objective of this systematic literature review is to identify, classify, compare and critically evaluate peer-reviewed research (2019–2026) on reinforcement learning for traffic signal control, with particular emphasis on RL formulations, evaluation environments and **validation realism**, reported traffic performance, deployment barriers and research gaps that separate simulation results from real-street use.

Specific objectives

#	Specific objective	Answered in
SO1	Quantify publication trends, venues, countries and application contexts	RQ1
SO2	Classify RL methods (algorithm family, single vs multi-agent, state, action, reward)	RQ2
SO3	Identify simulators, road networks, demand data and baselines used	RQ3
SO4	Code every study on the Validation Realism Level scale (VRL 0–5)	RQ3
SO5	Compare reported improvements in waiting time, queue length, travel time/delay and secondary outcomes	RQ4
SO6	Examine whether improvements depend on baseline strength and validation realism	RQ4

#	Specific objective	Answered in
SO7	Catalogue advantages, limitations and deployment barriers, and the solutions proposed	RQ5
SO8	Assess methodological quality and reproducibility	QA (supports all RQs)
SO9	Identify research gaps	RQ6
SO10	Recommend future research directions to move RL-TSC to real streets	RQ6

Step 5 — Research questions

The project proposal posed four questions; they are refined into six answerable RQs.

RQ	Question	Proposal question	Main extraction fields
RQ1 — Landscape	How has research on RL-based traffic signal control evolved between 2019 and 2026 in terms of volume, publication venues, countries and application contexts?	"how fast is this research area growing?"	year, venue, venue type, country, network scale, application context
RQ2 — Methods	Which RL formulations are used — algorithm family, agent architecture (single/multi-agent, centralised/decentralised, coordination mechanism), state, action and reward design — and how have they changed over time?	"Which RL methods are used…"	rl_family, algorithm, agent_architecture, coordination, state_repr, action_type, reward_type
RQ3 — Validation	How are RL-TSC methods evaluated (simulators, networks, demand data, baselines, robustness/transfer tests), and how are studies distributed across Validation Realism Levels 0–5?	"How are these methods tested: only in simulators, or also on real intersections?"	simulator, network_source, demand_source, baselines, vrl, transfer_test, robustness_test
RQ4 — Performance	What improvements in waiting time, queue length, travel time / delay (and throughput, stops, emissions, safety surrogates) are reported relative to fixed-time, actuated/Max-Pressure and RL baselines, and do they differ by baseline class and validation realism?	"How much do they actually improve traffic…?"	Quantitative evidence register (metric, RL value, baseline class, baseline value, relative improvement)
RQ5 — Strengths, limitations and barriers	What advantages, limitations and deployment barriers (sensing/data, sim-to-real gap, safety and regulatory compliance, robustness, computation/communication, interpretability, multimodality, scalability) are reported, and which mitigations are proposed?	"What stops them from being used in real cities…?"	barriers (coded), mitigations, limitations (authors' and reviewers')
RQ6 — Gaps and future research	Which scientific, methodological, technological and application gaps remain, and which approaches appear most promising for closing the simulation-to-street gap?	"…what should future research focus on?"	gap matrix, reviewer interpretation, future work statements

Validation Realism Level (VRL) — central coding scheme for RQ3/RQ4

VRL	Name	Definition (highest level reached by the study)
0	Synthetic simulation	Synthetic network (e.g., grid) and synthetic demand
1	Real-geometry simulation	Real road-network geometry, synthetic or uncalibrated demand

VRL	Name	Definition (highest level reached by the study)
2	Real-data simulation	Simulation driven by observed demand data (detector/camera/trajectory datasets) and/or calibrated against field measurements
3	Hardware-in-the-loop / live-data shadow mode	Real controller hardware or live sensor feeds in the loop, but the RL policy does **not** actuate street signals
4	Field test	RL policy actuates real signals at ≥ 1 real intersection during a limited trial
5	Sustained deployment	RL controller in continuous operational use (≥ 1 month) with reported outcomes

Supplementary flags (independent of VRL): `transfer_test` (evaluated on unseen scenarios/simulators), `robustness_test` (detector noise/failure, demand surges, incidents), `safety_constraints` (min green, pedestrian clearance, action masking).

Document 06 — Systematic Review Protocol (Step 6)

Field	Value
Protocol version	**1.0**
Date	17 Oct 2026
Reporting framework	PRISMA 2020 (27-item checklist + abstract checklist + flow diagram, new review, databases + other sources); PRISMA-S for search reporting
Registration	Planned: OSF Registries (open, free). Record the OSF DOI here once registered.
Reviewers	**R1** — Rakhimzhanov Amangeldy; **R2** — Nurkin Aidar

1. Title

Smart Traffic Lights That Learn: A Systematic Literature Review of Reinforcement Learning for Traffic Signal Control — From Simulation to Real Streets.

2. Objectives and research questions

See Document 04–05 (main objective, SO1–SO10, RQ1–RQ6, VRL scale).

3. Information sources (databases)

Scopus, Web of Science Core Collection, IEEE Xplore (core). Other sources: backward/forward citation chaining (Google Scholar, publisher pages), recorded separately as “other methods” in PRISMA. Details: Document 08.

4. Search period and strategy

1 Jan 2019 – final search date. Two-concept Boolean strategy (RL × traffic signal) searched in title, abstract and keywords. Database-specific strings: Document 09. Pilot: Document 10. Locked version: Document 11. Multi-wave plan: Document 12.

5. Eligibility criteria

Inclusion criteria (all must be met)

Code	Criterion
IC1	Primary study that proposes, adapts or evaluates an **RL-based controller** for **road traffic signals** (phase selection, phase duration, cycle, split or offset) at ≥ 1 intersection
IC2	Reports an **empirical evaluation** (simulation, hardware-in-the-loop or field) with ≥ 1 **quantitative traffic performance outcome**
IC3	Published 1 Jan 2019 – final search date (online first date counts)

Code	Criterion
IC4	Peer-reviewed **journal article** or **full conference paper**
IC5	Written in **English**
IC6	Full text retrievable by the review team (institutional access, open access, interlibrary loan or author request)

Exclusion criteria (one primary reason is recorded per excluded full text)

Code	Reason
E1	Wrong application — not road traffic signal control (ramp metering only, routing, autonomous driving, power/telecom "signals", etc.)
E2	Wrong method — RL is not the evaluated controller (non-RL optimisation, prediction only, prompting-only LLM, RL only as minor baseline)
E3	Wrong publication type — review, survey, editorial, position paper, book chapter
E4	No empirical evaluation or no quantitative traffic outcome
E5	Outside date range
E6	Not peer-reviewed (preprint, thesis, report, poster, extended abstract < 4 pages)
E7	Not in English
E8	Duplicate record not detected at deduplication (same report)
E9	Full text not retrievable after documented attempts (IC6)
E10	Insufficient technical information to identify the RL formulation or the results

The order E1 → E10 is the **hierarchy**: when several reasons apply, the first applicable code is recorded.

6. Study selection procedure

1. **Deduplication** (Step 14): automatic merge on DOI and on exact normalised title + year + first-author surname; fuzzy-title matches (similarity ≥ 0.93) are *flagged* and confirmed manually. Tool: `python -m slr dedup`.
2. **Title/abstract screening** (Step 15)
    - Calibration: R1 and R2 independently screen a random 10 % sample (minimum 50 records); Cohen's κ is computed (`python -m slr screen-ta --stats`). Target κ ≥ 0.60; otherwise discuss, clarify criteria (logged as protocol clarification) and repeat with a new sample.
    - Remaining records are split between R1 and R2. Every record a reviewer marks *Exclude – low confidence* is checked by the other reviewer.
    - Decisions: Include / Exclude / Uncertain. **Conservative rule:** Include or Uncertain from either reviewer → full-text stage.
    - A keyword relevance score is shown only to sort the list; it never makes a decision.
3. **Full-text eligibility** (Step 16): both reviewers assess **all** full texts independently; one primary exclusion code per excluded report; disagreements resolved by discussion, recorded in the `notes` field.
4. **Reports vs studies:** a journal extension of a conference paper reporting the same experiments is kept as a *report* but linked through a **Study Family ID** (Step 27), so it is not double counted in the synthesis.

7. Quality assessment (Step 18)

Five domains scored 0 (No) / 1 (Partial) / 2 (Yes), maximum 10. Thresholds fixed **a priori**: High 8–10, Moderate 5–7, Limited 0–4. QA is **not** an exclusion criterion; it is used for interpretation and sensitivity analysis. Both reviewers score all included studies; κ (weighted) is reported. Full rubric: Document 18.

8. Data extraction (Steps 19–21)

Codebook with controlled vocabularies: Document 19–21 and `data/protocol/codebook.csv`. Two evidence tables: Technical extraction (one row per report) and Quantitative evidence register (one row per number, with source page/table/figure). R1 extracts, R2 verifies 100 % of quantitative entries and a 20 % random sample of technical entries.

9. Synthesis plan (Steps 22–27)

- Taxonomy of methods and validation (Step 22); method-family comparison table (Step 23).
- Descriptive bibliometrics (Step 24).
- Quantitative synthesis of **relative improvement** RI = (baseline − RL)/baseline × 100 % for lower-is-better metrics (and (RL − baseline)/baseline × 100 % for higher-is-better), stratified by metric × baseline class × VRL; median, IQR, min, max, mean, SD (Step 25).
- Meta-analysis only if the pre-specified feasibility rule (Document 26) is met; otherwise structured narrative synthesis with subgroup comparison.
- Dependence control: Study Family ID and Dataset/Scenario ID; summary statistics computed at family level (Step 27).
- Sensitivity analysis: excluding Limited-quality studies; excluding families that contribute > 10 % of entries.

10. Reviewer responsibilities

Task	R1 (Amangeldy)	R2 (Aidar)
Search execution & export	Scopus, IEEE Xplore	Web of Science, citation chaining
Search log / PRISMA-S	Owner	Verifier
Title/abstract screening	Calibration + 50 %	Calibration + 50 %
Full-text screening	100 %	100 %
QA	100 %	100 %
Technical extraction	Owner	20 % verification
Quantitative extraction	Owner	100 % verification
Synthesis & figures	Shared	Shared
Disagreements	Discussion; unresolved → decided by consensus with supervisor	

11. Tools

Python pipeline in this repository (`slr/`), master evidence workbook (`outputs/Master_Evidence_Register.xlsx`), reference manager (Zotero) for PDFs.

12. Amendments log

Any change to scope, search concepts, eligibility, RQs or QA after the search is locked is recorded here (and in `data/protocol/amendments.csv`) with date, reason and impact.

#	Date	Section	Change	Reason	Impact
—	—	—	(none yet)	—	—

Part II — Search strategy (Steps 7–12)

Document 07 — Search Concept Matrix (Step 7)

The topic is split into concepts. Synonyms are joined with OR inside a concept; the concepts are joined with AND.

Concept	Role	Terms (OR-joined)	Used in
A — Reinforcement learning	Technology	"reinforcement learning" · "Q-learning" · "deep Q-network(s)" · DQN · "actor-critic" · "policy gradient" · "proximal policy optimization/optimisation"	Core (Wave 1)
B — Traffic signal control	Application	"traffic signal(s)" · "traffic light(s)" · "signalized/signalised intersection(s)" · "signal timing" · "signal phase(s)/phasing" · "adaptive signal control"	Core (Wave 1)
C — Deployment / realism	Target phenomenon (precision/targeting only)	"field test" · "field trial" · "real-world deployment" · deploy* · "sim-to-real" · sim2real · "hardware-in-the-loop" · "digital twin" · "real intersection" · "transfer learning" · "domain randomization"	Wave 2 & Wave 4 targeted searches
D — Expansion terms found during Wave 1	Terminology expansion	SARSA · "temporal difference" · DDPG · "soft actor-critic" · A2C/A3C · MARL · "multi-agent learning" · "learning-based signal" · "traffic light controller" · "intersection controller" · "phase selection"	Wave 2
E — Context (optional)	Context	urban · arterial · corridor · network · pedestrian · "public transport" · "emergency vehicle" · "connected vehicle"	Wave 4 gap-targeted

Combination logic

- **Core query (Wave 1):** `(A) AND (B)` in title/abstract/keywords, 2019–2026, articles + conference papers, English.
- Concept C is **not** used in the core query because it would remove the large simulation-only literature that is needed to answer RQ3/RQ4 (the gap itself is the object of study). It is used only to *add* recall for deployment studies (Wave 2/4).

Design notes

- "signal control" alone was rejected: in the pilot it retrieves power-electronics and telecommunications records (low precision).
- "intersection" alone was rejected: too broad (vehicle intersection management without signals, set intersection in maths).
- Truncation is used where the database supports it inside phrases (Scopus, WoS); IEEE Xplore variants are written out in full (see Document 09).

Document 08 — Database Selection (Step 8)

Database	Platform	Coverage relevant to RL-TSC	Role
Scopus	Elsevier, scopus.com	Broadest multidisciplinary coverage; indexes ACM (KDD, CIKM), AAAI, IEEE, Elsevier (TR-C, ESWA), Springer, MDPI, Taylor & Francis	Core 1 (highest metadata priority in deduplication)
Web of Science Core Collection	Clarivate, webofscience.com — SCI-EXPANDED, SSCI, ESCI, CPCI-S	Independent curated index; strong for transport journals	Core 2

Database	Platform	Coverage relevant to RL-TSC	Role
IEEE Xplore	IEEE, ieeexplore.ieee.org	Domain-specific: IEEE T-ITS, T-VT, IoT-J, ITSC, CDC, CASE, SMC — many early-access papers not yet in Scopus/WoS	Core 3 (domain-specific)
Google Scholar	scholar.google.com	Forward citation chaining ("Cited by")	Other methods (Wave 3) — not a PRISMA database
ACM Digital Library	dl.acm.org	KDD/CIKM proceedings — cross-check during pilot only	Pilot check

Principle followed: **three independent bibliographic databases** (two multidisciplinary + one domain-specific), as recommended for engineering SLRs (Scopus + WoS + domain database).

Not used, with reason

- **arXiv** — preprints are excluded (E6); arXiv versions are only used to locate the peer-reviewed version.
- **PubMed / Embase** — not relevant to traffic engineering.
- **TRID (TRB)** — considered; excluded because export is limited and coverage of RL papers overlaps Scopus/WoS. Documented as a limitation.

Document 09 — Database-Specific Search Strings (Step 9)

Each string implements `(Concept A) AND (Concept B)` from Document 07. The exact strings run, the date of the run and the number retrieved **must be recorded in `data/protocol/search_log.csv`** (PRISMA-S items 1–16). Copy the strings exactly; do not retype them.

S1 — Scopus (Advanced search)

Fields: `TITLE-ABS-KEY` (title, abstract, author keywords, index keywords).

```
TITLE-ABS-KEY ( ( "reinforcement learning" OR "Q-learning" OR "deep Q-network*" OR "DQN" OR "actor-critic"
OR "policy gradient" OR "proximal policy optimi*" ) AND ( "traffic signal*" OR "traffic light*" OR "signali?
ed intersection*" OR "signal timing" OR "signal phas*" OR "adaptive signal control" ) ) AND PUBYEAR > 2018
AND PUBYEAR < 2027 AND ( LIMIT-TO ( DOCTYPE , "ar" ) OR LIMIT-TO ( DOCTYPE , "cp" ) ) AND ( LIMIT-TO (
LANGUAGE , "English" ) )
```

Export: *Export → CSV*, select **all** records and these field groups: Citation information, Bibliographical information, Abstract & keywords (include EID). Save as `data/raw/scopus/scopus_wave1_YYYYMMDD.csv`. Scopus allows 20 000 records per CSV export; if more, split by year.

S2 — Web of Science Core Collection (Advanced search)

Editions: SCI-EXPANDED, SSCI, ESCI, CPCI-S. Field: `TS` (topic = title, abstract, author keywords, Keywords Plus).

```
TS=( ("reinforcement learning" OR "Q-learning" OR "deep Q-network*" OR DQN OR "actor-critic" OR "policy
gradient" OR "proximal policy optimi*") AND ("traffic signal*" OR "traffic light*" OR "signali?ed
intersection*" OR "signal timing" OR "signal phas*" OR "adaptive signal control") ) AND PY=(2019-2026)
```

Refine after running: Document Types = *Article*, *Proceedings Paper*, *Early Access*; Languages = *English*. Export: *Export → Tab delimited file*, record content **Full Record**, in batches of 1 000 (`savedrecs.txt`, `savedrecs(1).txt`, …). Save as `data/raw/wos/wos_wave1_YYYYMMDD_partN.txt`.

S3 — IEEE Xplore (Command Search)

Field: `All Metadata`. IEEE Xplore limits wildcards, so plural/spelling variants are written out.

```
("All Metadata":"reinforcement learning" OR "All Metadata":"Q-learning" OR "All Metadata":"deep Q-network"
OR "All Metadata":"DQN" OR "All Metadata":"actor-critic" OR "All Metadata":"policy gradient" OR "All
Metadata":"proximal policy optimization") AND ("All Metadata":"traffic signal" OR "All Metadata":"traffic
signals" OR "All Metadata":"traffic light" OR "All Metadata":"traffic lights" OR "All Metadata":"signalized
intersection" OR "All Metadata":"signalized intersections" OR "All Metadata":"signalised intersection" OR
"All Metadata":"signal timing" OR "All Metadata":"signal phase" OR "All Metadata":"adaptive signal control")
```

Filters: Year 2019–2026; Content type = *Conferences*, *Journals*, *Early Access Articles*. Export: *Export → Search Results → CSV* (max 2 000 per export; split by year if needed). Save as `data/raw/ieee/ieee_wave1_YYYYMMDD.csv`.

> IEEE "All Metadata" covers title, abstract, author keywords and IEEE index terms, but not the full text. It is therefore comparable to TITLE-ABS-KEY / TS. If the pilot shows a precision problem, restrict the search to the `"Document Title"`, `"Abstract"` and `"Author Keywords"` fields and record the change as a pilot iteration.

Search log template (PRISMA-S)

Fill one row per executed search in `data/protocol/search_log.csv`:

search_id	wave	database	platform	date_run (DD/MM/YYYY)	fields	query (exact)	filters	records_retrieved	export_file	run_by	notes
S1	1	Scopus	scopus.com		TITLE-ABS-KEY	(string S1)	2019–2026; ar, cp; English		data/raw/scopus/…	R1	
S2	1	Web of Science CC	webofscience.com		TS	(string S2)	2019–2026; Article, Proc. Paper, Early Access; English		data/raw/wos/…	R2	
S3	1	IEEE Xplore	ieeexplore.ieee.org		All Metadata	(string S3)	2019–2026; Conf., Journals, Early Access		data/raw/ieee/…	R1	

`python -m slr ingest` compares the number of records parsed from each export with `records_retrieved` and warns about any mismatch.

Document 10 — Pilot Search Procedure (Step 10)

Purpose

Check **recall** (does the search find papers we already know are relevant?) and **precision** (how much noise?) before locking the strategy.

Benchmark ("gold") set

`data/protocol/known_relevant_papers.csv` lists 20 papers identified during scoping (Document 02) that any adequate search must retrieve. It deliberately mixes:

- landmark DRL methods (PressLight, CoLight, FRAP, MPLight, MetaLight, AttendLight, MA2C, Advanced-XLight, CoSLight);
- deployment/realism papers (Müller et al. 2021; Müller & Sabatelli 2022; Egea et al. 2020; UGAT 2023; PromptGAT 2024);
- benchmark and LLM papers (RESCO 2021, LibSignal, LLMLight);

- reviews (Noaeen 2022; Haydari & Yilmaz 2022; Michailidis 2025) — retrieved by the search but later excluded with E3; they are needed for Wave 3 chaining.

Some gold papers appear in venues that are not indexed by every database (e.g., NeurIPS). Recall is therefore assessed **across the union of the three databases**, and per database for information.

Procedure

1. Run S1–S3 (Document 09) and export results into `data/raw/<db>/` with file names starting `pilot_`.
2. Run `python -m slr pilot --raw-glob "data/raw/*/pilot_*"`. The tool matches gold papers to the exports by DOI or normalised title and writes `outputs/pilot/pilot_recall.csv` plus a summary.
3. **Recall criterion:** ≥ 90 % of the gold papers that are indexed in at least one of the databases must be retrieved by the union of searches. For each miss, record *why* (not indexed / terminology not covered / filter problem).
4. **Precision check:** screen a random sample of 100 retrieved titles (`python -m slr pilot --precision-sample 100` writes the sample). Report the proportion that is plausibly relevant.
5. If a miss is caused by terminology, add the missing synonym to Concept A/B (Document 07), re-run and repeat. If precision < 20 %, consider adding context terms — but never tune the search towards preferred papers only.
6. Record every iteration in `data/protocol/pilot_log.csv` (iteration, change, recall, precision, decision).

Result

To be completed after the pilot is run:

Iteration	Change	Gold retrieved (union)	Recall	Precision (sample)	Decision
1	Strings S1–S3 v0.9				

The pilot exports are **not** used as review evidence; Wave 1 is re-run after locking.

Document 11 — SEARCH PROTOCOL VERSION 1.0 — LOCKED (Step 11)

> Locking takes effect when the pilot (Document 10) meets its recall criterion and both reviewers sign below. After that, any change is an **amendment** (Document 06 §12, `data/protocol/amendments.csv`).

Element	Locked value
Databases	Scopus; Web of Science Core Collection (SCI-EXPANDED, SSCI, ESCI, CPCI-S); IEEE Xplore
Strings	S1, S2, S3 exactly as in Document 09 (machine-readable copy: `data/protocol/search_strings.json`)
Fields	Scopus TITLE-ABS-KEY; WoS TS; IEEE All Metadata
Period	2019-01-01 to date of final search
Document types	Journal article (incl. early access), conference paper
Language	English
Inclusion criteria	IC1–IC6 (Document 06 §5)
Exclusion criteria	E1–E10, applied in hierarchical order (Document 06 §5)
QA	QA1–QA5, 0/1/2, thresholds 8–10 / 5–7 / 0–4 (Document 18)
Extraction	Codebook v1.0 (`data/protocol/codebook.csv`)

Machine-readable configuration used by the pipeline: `slr/config.py`.

Role	Name	Date	Signature
Reviewer 1	Rakhimzhanov Amangeldy		
Reviewer 2	Nurkin Aidar		

Document 12 — Multi-Wave Search Strategy (Step 12)

Every wave is logged in `data/protocol/search_log.csv` (column `wave`) and its exports are stored in `data/raw/<source>/` with the wave in the file name (`…_wave2_…`). The pipeline reads the wave from the file name and reports PRISMA counts per source and wave.

Wave	Purpose	How	PRISMA box
1 — Core database search	Main evidence base	Run locked S1–S3 in all three databases	Records from databases
2 — Terminology expansion	Recall for synonyms found in Wave 1 (Concept D) and deployment terms (Concept C)	Strings S1b–S3b = `(A ∪ D) AND (B)` *NOT* already-retrieved results; and `(A ∪ D) AND (B) AND (C)`; same filters	Records from databases (wave 2)
3 — Citation chaining	Find studies missed by keywords	**Backward:** reference lists of (a) all included studies with QA ≥ 8 and (b) the reviews in Document 02. **Forward:** Google Scholar "Cited by" for the same set. Candidates entered in `data/raw/other/citation_chaining.csv` (unified format)	Records from other methods: citation searching
4 — Gap-targeted and update search	(a) Under-represented contexts found in the gap matrix (e.g., pedestrians, public transport, field tests); (b) update to capture papers published after Wave 1	(a) `(A) AND (B) AND (E-term)`; (b) re-run S1–S3 restricted to the load date ≥ Wave 1 date, immediately before submission	(a) databases; (b) databases — reported as update

Rules

1. A record found in several waves is deduplicated like any other; its first wave is kept as `first_wave`.
2. Supplementary searches never change eligibility criteria.
3. Citation-chaining candidates are screened with the same two-stage process.
4. The date of the last search (Wave 4b) is the review's "search date" reported in the abstract (PRISMA item 7).

Unified CSV columns for `data/raw/other/*.csv`: `source, title, authors, year, venue, doi, abstract, keywords, doc_type, language, url, found_via`

Part III — Record management and PRISMA (Steps 13–17)

Documents 13–17 — Record Management, Screening and PRISMA (Steps 13–17)

All commands are run from the repository root. `python -m slr all` runs every stage that has input.

Step 13 — Master search register

The register has **CSV files as the source of truth** (versioned in git) and an Excel snapshot regenerated on every run:

Workbook sheet	Source file	Who edits
Search_Log	`data/protocol/search_log.csv`	Reviewers
Raw_Records	`data/records/raw_records.csv`	Pipeline
Deduplication	`data/records/dedup_log.csv` (+ `dedup_candidates.csv`)	Pipeline (+ reviewers confirm fuzzy pairs)
Title_Abstract_Screening	`data/screening/title_abstract.csv`	Reviewers (decision columns)
Full_Text_Screening	`data/screening/full_text.csv`	Reviewers (decision columns)
Included_Studies	`data/screening/included_studies.csv`	Pipeline
Excluded_Full_Text	derived from full_text.csv	Pipeline
Quality_Assessment	`data/quality/quality_assessment.csv`	Reviewers (score columns)
Technical_Extraction	`data/extraction/technical_extraction.csv`	Reviewers
Quantitative_Extraction	`data/extraction/quantitative_extraction.csv`	Reviewers
RQ_Evidence	`data/synthesis/rq_evidence.csv`	Reviewers
Research_Gaps	`data/synthesis/research_gaps.csv`	Reviewers
PRISMA_Counts	`outputs/prisma/prisma_counts.csv`	Pipeline
Reference_Audit	`data/protocol/reference_audit_template.csv`	Reviewers (later steps)
Submission_Audit	`data/protocol/submission_audit_template.csv`	Reviewers (later steps)

Build it: `python -m slr workbook` → `outputs/Master_Evidence_Register.xlsx`.

Golden rule: reviewers only type into the decision/score/extraction columns. Re-running any command keeps that input (merged by `record_id` / `paper_id`). Rows that disappear are saved to `*_orphans.csv`, so a decision is never lost silently.

Workflow after the search is locked

1. Save exports in `data/raw/scopus/`, `data/raw/wos/`, `data/raw/ieee/` (and `data/raw/other/` for Wave 3), wave number in the file name (e.g. `scopus_wave1_20261020.csv`).
2. Fill one row per executed search in `data/protocol/search_log.csv` (date, exact string, filters, `records_retrieved`, `export_file`).
3. `python -m slr ingest` → parses all exports; warns if the parsed count ≠ `records_retrieved`.

Step 14 — Deduplication

`python -m slr dedup`

- Automatic merge only when it is safe: same DOI, or identical normalised title + year ±1 + same first-author surname **without** conflicting DOIs.
- Similar titles (≥ 0.93) are **not** merged automatically. They are listed in `data/records/dedup_candidates.csv`; set `is_duplicate` = `Y` or `N` and re-run `dedup`. A conference paper and its journal extension usually have the same title but different DOIs. Mark these `N`: they are different *reports*, and they are linked later through a study family (Step 27).
- Metadata priority when merging: Scopus > WoS > IEEE > other. Empty fields (e.g. abstract) are filled from the other copies.

- Protocol filters (year, document type, language) are applied **after** deduplication and reported as "records removed for other reasons".
- The balance is checked by the tests: identified − duplicates − removed = screened.

Step 15 — Title/abstract screening

`python -m slr screen-ta` creates `data/screening/title_abstract.csv`:

Column	Meaning
`assigned_to`	R1+R2 = calibration sample (10 %, min 50, random seed 2026); otherwise R1 or R2 (deterministic 50/50 split)
`relevance_score`	Keyword score used **only to sort**; never decides anything
`r1_decision`, `r2_decision`	`include` / `exclude` / `uncertain` (or I / E / U)
`r1_confidence`, `r2_confidence`	`low` for an uncertain exclusion; it then needs the other reviewer
`override_decision`	Consensus after discussion (optional)
`final_decision`	Computed with the conservative rule: include/uncertain from either reviewer → full text

1. Both reviewers screen the calibration rows first, then run `python -m slr screen-ta --stats`. If κ < 0.60, discuss, clarify the criteria (record the clarification in the protocol amendments), and repeat with a new sample.
2. Screen your assigned rows. Mark any exclusion you are unsure about as `low`.
3. Re-run `python -m slr screen-ta` until no `pending` rows remain.

Step 16 — Full-text eligibility

`python -m slr screen-ft` creates `data/screening/full_text.csv` for every record with final decision include/uncertain.

- `retrieved` = `Y` or `N`; if `N`, write the attempts made in `retrieval_note` (institutional access, open-access search, interlibrary loan, e-mail to the author).
- `r1_decision` and `r2_decision` = `include` / `exclude`. Every exclusion needs **exactly one** code E1–E10 (Protocol §5) in `r1_reason` / `r2_reason`.
- If the decisions disagree, the final decision is `conflict`. Discuss and fill `consensus_decision` (and `consensus_reason`).
- If both reviewers exclude but give different codes, the hierarchy (first applicable code, E1 → E10) decides automatically.
- Included reports receive stable IDs (`P001`, …) in `data/screening/included_studies.csv`.

Step 17 — PRISMA flow diagram

`python -m slr prisma` → `outputs/prisma/prisma_flow.png|svg` and `prisma_counts.csv`.

- All numbers are computed from the files above, never typed in.
- Database counts are reported per database. Citation-searching records are reported in the separate "other methods" column.
- "Studies included" = number of distinct study families. "Reports" = number of included papers.
- While any stage is incomplete, the diagram carries a red **DRAFT** banner that names what is pending.

Part IV — Quality assessment (Step 18)

Document 18 — Quality Assessment Framework (Step 18)

The five generic domains of the guideline, adapted to RL traffic-signal-control studies. Each item is scored **0 = No, 1 = Partial, 2 = Yes**; maximum **10**.

Item	Domain	2 = Yes	1 = Partial	0 = No
QA1	Research design	Objective stated **and** full MDP formulation given: state, action, reward, agent architecture (and, for MARL, coordination)	Objective clear, but one of state/action/reward is vague	Objective or RL formulation unclear
QA2	Data quality	Real road network **and** real or calibrated demand, fully described (source, period, volumes)	Synthetic network/demand that is fully described (reproducible), **or** real data that is poorly described	Network/demand not described
QA3	Reproducibility	Hyperparameters, training procedure and simulator/version reported **and** code or scenario files publicly available	Details reported but no code/scenario files	Insufficient details to re-implement
QA4	Validation	Compared with ≥ 1 non-RL adaptive baseline (actuated / Max-Pressure / SOTL / SCATS-SCOOT-like) **and** (a state-of-the-art RL baseline **or** a test on unseen scenarios **or** a field/HIL test)	Only fixed-time, or only one baseline, or only training scenarios	No baseline
QA5	Results reporting	Results over multiple runs/seeds **with** dispersion (SD/CI) **and** limitations discussed	One of the two	Neither

Thresholds (fixed a priori — do not change after seeing results)

Total	Class
8–10	High
5–7	Moderate
0–4	Limited

Use of QA

- QA is **not** an exclusion criterion.
- Uses: (1) the interpretation in the Discussion section; (2) a sensitivity analysis that excludes Limited studies (`sensitivity_excl_limited_qa.csv`); (3) choosing the seed set for Wave 3 citation chaining (QA ≥ 8).

Procedure

1. `python -m slr qa` creates `data/quality/quality_assessment.csv` (one row per included paper).
2. R1 fills `r1_QA1`…`r1_QA5`, R2 fills `r2_QA1`…`r2_QA5`, independently.
3. Re-run `python -m slr qa`. Items where the reviewers disagree are listed in `qa_conflicts`. Discuss them and fill `consensus_QAx`.
4. Report the linear weighted κ for each item (printed by the command) in the Methods section.

Part V — Data extraction (Steps 19–21)

Documents 19–21 — Data Extraction (Steps 19–21)

Step 19 — Extraction schema (designed before extraction)

The codebook is defined once in `slr/codebook.py` and exported to `data/protocol/codebook.csv`, with the field, guideline section A–H, supported RQ, type, allowed values and a description. Controlled vocabularies are validated automatically: a misspelt value is reported every time `python -m slr extract` runs.

Guideline section	Fields (technical table)	RQ
A Bibliographic	paper_id, authors, year, title, venue, venue_type, doi, country_first_author, study_location	RQ1
B Application	application_context, n_intersections, traffic_modes, objective_focus	RQ1, RQ6
C Technology / RL formulation	rl_family, algorithm, agent_architecture, coordination, state_repr, action_type, reward_type, function_approximator, simulator, sensing_source	RQ2, RQ3
D/F Methodology & validation	network_source, demand_source, dataset_ids, baseline_classes, **vrl**, transfer_test, robustness_test, safety_constraints, n_seeds, code_available	RQ3, RQ4
E Quantitative results	separate table (Step 21)	RQ4
G Conclusions	key_findings, advantages, limitations_authors	RQ4, RQ5
H Reviewer interpretation	barriers, mitigations, reviewer_strength, reviewer_weakness, research_gap_note	RQ5, RQ6
Dependence	study_family_id, campaign_id	Step 27

Coding rules

- **VRL** is the *highest* level the study reaches (definitions in Document 04–05). A study with a calibrated simulation and a two-week field test is VRL 4.
- **dataset_ids** use canonical names so that reuse can be counted: `jinan_3x4`, `hangzhou_4x4`, `ny_16x3`, `ny_28x7`, `manhattan_large`, `shanghai`, `monaco`, `ingolstadt1/7/21`, `cologne1/3/8`, `atlanta`, `synthetic_grid_NxM`. Add new names to this list when you first use them.
- `barriers` records what the paper **discusses or evidently faces**, e.g. a camera-based state with no noise test → `sensing_data`.
- Pilot the form on 5 papers (both reviewers), compare, refine the descriptions (not the categories), then extract the rest.

Step 20 — Traceable evidence register

Every number goes into `data/extraction/quantitative_extraction.csv` as **one row per value pair**:

entry_id	paper_id	scenario_id	metric	rl_value	baseline_name	baseline_class	baseline_value	unit	source_location
Q0001	P014	hangzhou_4x4	travel_time	287.3	MaxPressure	max_pressure	312.1	s	Table 4, p. 9

- `entry_id` is assigned automatically. `source_location` is **mandatory**, and the pipeline lists every entry that has none.
- When a paper reports several experiments, use one row per experiment and describe the condition in `context` (demand level, peak/off-peak). Never average across experiments yourself.
- When the paper gives only a % improvement, put it in `reported_improvement_pct` and leave the raw values empty.
- Values read off a figure: add "(read from figure)" in `context`.
- R2 verifies **100 %** of the quantitative entries (`verified_by`).

Step 21 — Two separate evidence tables

- **Technical evidence** — `technical_extraction.csv`, one row per paper. Pre-filled with the bibliographic data and a guess of the first author's country; check the guess.

- **Quantitative evidence** — `quantitative_extraction.csv`, one row per number. The pipeline adds:
  - `direction` (lower or higher is better, from the metric);
  - `relative_improvement_pct` = (B − RL)/B × 100 for lower-is-better metrics, or (RL − B)/B × 100 for higher-is-better metrics. Positive always means RL is better;
  - `study_family_id` and `vrl_effective`, inherited from the technical table.

Commands: `python -m slr extract` (creates or updates both sheets, validates them and computes the derived columns).

Part VI — Classification and synthesis (Steps 22–27)

Documents 22–27 — Classification and Synthesis (Steps 22–27)

`python -m slr synthesis` computes everything below from the extraction sheets and writes:

- `outputs/synthesis/synthesis_report.md` — all results in one place, with tables and figure links;
- `outputs/synthesis/tables/*.csv` — every table, ready for the manuscript;
- `outputs/synthesis/figures/*.png` — 200-dpi figures;
- `outputs/synthesis/taxonomy.md` — the taxonomy with counts and a Mermaid mind map.

Step 22 — Taxonomy

Hierarchy (preliminary, built from scoping; the counts are filled from the data):

```
RL for traffic signal control
├── Application context: isolated · arterial corridor · grid · real urban network
├── Road users: private vehicles · pedestrians · cyclists · public transport · emergency · CAV
├── RL formulation
│   ├── Family: tabular · value-based deep · policy gradient · actor–critic · model-based · hierarchical · meta-RL · offline RL · LLM–RL hybrid
│   ├── Agents: single · independent MARL · cooperative MARL · centralised
│   ├── Coordination: none · neighbour observation · communication · graph attention · pressure-based · hierarchical
│   ├── State / Action / Reward
│   └── Function approximator: table · MLP · CNN · RNN · GNN · attention · LLM
├── Evaluation: simulator · network source · demand source · baselines
├── Validation realism: VRL0 → VRL5 (+ transfer, robustness, safety flags)
├── Performance indicators: travel time · waiting time · delay · queue · throughput · stops · emissions · safety
├── Deployment barriers: sensing · sim-to-real · safety/compliance · robustness · computation · communication · interpretability · scalability · multimodality · training cost · regulation
└── Research gaps (gap matrix)
```

The final taxonomy figure for the manuscript is drawn from `taxonomy.md`.

Step 23 — Comparison of methodological families

`tables/method_family_comparison.csv` combines:

- **computed** columns: number of papers, top algorithms, median and maximum VRL, % with transfer, robustness and code-availability tests, median RI against fixed-time, Max-Pressure and RL baselines, top barriers;
- **conceptual** columns (`data/protocol/method_families.csv`): principle, advantages, limitations, typical validation, best application. Revise these after extraction so that they reflect the evidence.

Step 24 — Bibliometric and descriptive analysis

Papers per year (split by journal/conference), top venues, countries, application context, road users, objectives, RL family over time, algorithms, agent architecture, coordination, state, action, reward, approximator, simulators, network and demand sources, baselines, **VRL distribution**, validation flags, barriers, QA classes. Heat map: RL family × VRL.

Step 25 — Quantitative synthesis

- Effect measure: relative improvement RI (%), positive = RL better.
- Stratification: metric × baseline class (primary), metric × VRL (to test whether gains shrink as validation becomes more realistic), and a sensitivity analysis that excludes Limited-QA studies.
- Statistics: n values, n papers, n families, min, Q1, median, Q3, max, mean, SD.

- **Primary level = study family**: one value per family and cell, the median of its entries. Entry-level statistics are reported as a secondary analysis.
- Figures: box plots (RI by baseline class, one panel per metric, with every family shown as a point) and a scatter of RI against VRL.
- Different metrics are never pooled together; the relative scale makes units comparable only *within* a metric.

Step 26 — Is a meta-analysis appropriate?

Pre-specified rule: a cell (metric × baseline class) is eligible only if **≥ 5 independent study families** report mean, SD and number of runs, all in the same unit. For eligible cells, `meta_analysis_feasibility.csv` gives an *exploratory* random-effects (DerSimonian–Laird) pooled log response ratio, converted back to %, with a 95 % CI, τ^2 and I^2.

Expected outcome, based on scoping: most cells will **not** be eligible (dispersion is rarely reported and scenarios are heterogeneous). In that case the review uses a structured narrative synthesis with subgroup comparison and descriptive statistics, and says so explicitly. A meta-analysis is not forced.

Step 27 — Dependence between publications

- `study_family_id`: reports of the same experiments, project or code base share an ID (e.g. a conference paper and its journal extension; a series that reuses the same trained agents).
- `campaign_id`: the same field trial.
- `dataset_ids` / `scenario_id`: the same benchmark dataset. Reuse is counted in `dataset_reuse.csv`. Shared *public* datasets (e.g. Jinan, Hangzhou) do **not** by themselves make studies dependent, but the reuse is reported because it limits external validity.
- `study_families.csv` flags families that contribute **> 10 %** of the quantitative entries (`dominant_family_gt10pct`). These are checked in the sensitivity analysis.
- PRISMA reports both *studies* (families) and *reports* (papers).

Appendices — protocol data files

Appendix A — Pilot benchmark (gold) set

Source file: `data/protocol/known_relevant_papers.csv`

gold_id	title	first_author	year	venue	role
G01	PressLight: Learning Max Pressure Control to Coordinate Traffic Signals in Arterial Network	Wei	2019	KDD 2019	landmark method
G02	CoLight: Learning Network-level Cooperation for Traffic Signal Control	Wei	2019	CIKM 2019	landmark method
G03	Learning Phase Competition for Traffic Signal Control	Zheng	2019	CIKM 2019	landmark method (FRAP)
G04	Toward A Thousand Lights: Decentralized Deep Reinforcement Learning for Large-Scale Traffic Signal Control	Chen	2020	AAAI 2020	landmark method (MPLight)
G05	Multi-Agent Deep Reinforcement Learning for Large-Scale Traffic Signal Control	Chu	2020	IEEE Transactions on Intelligent Transportation Systems	landmark method (MA2C)
G06	MetaLight: Value-Based Meta-Reinforcement Learning for Traffic Signal Control	Zang	2020	AAAI 2020	meta-RL
G07	AttendLight: Universal Attention-Based Reinforcement Learning Model for Traffic Signal Control	OroojIooy	2020	NeurIPS 2020	universal model
G08	Expression might be enough: representing pressure and demand for reinforcement learning based traffic signal control	Zhang	2022	ICML 2022	pressure-based state
G09	CoSLight: Co-optimizing Collaborator Selection and Decision-making to Enhance Traffic Signal Control	Ruan	2024	KDD 2024	graph/collaboration
G10	Towards Real-World Deployment of Reinforcement Learning for Traffic Signal Control	Müller	2021	ICMLA 2021	deployment
G11	Safe and Psychologically Pleasant Traffic Signal Control with Reinforcement Learning using Action Masking	Müller	2022	IEEE ITSC 2022	safety/deployment

gold_id	title	first_author	year	venue	role
G12	Assessment of Reward Functions for Reinforcement Learning Traffic Signal Control under Real-World Limitations	Egea	2020	IEEE SMC 2020	real-world limitations
G13	Uncertainty-aware Grounded Action Transformation towards Sim-to-Real Transfer for Traffic Signal Control	Da	2023	IEEE CDC 2023	sim-to-real
G14	Prompt to Transfer: Sim-to-Real Transfer for Traffic Signal Control with Prompt Learning	Da	2024	AAAI 2024	sim-to-real
G15	Sim2Real Transfer for Traffic Signal Control	Da	2023	IEEE CASE 2023	sim-to-real
G16	LLMLight: Large Language Models as Traffic Signal Control Agents	Lai	2025	KDD 2025	LLM + RL hybrid
G17	LibSignal: An Open Library for Traffic Signal Control	Mei	2023	Machine Learning (journal) / NeurIPS 2022 workshop	benchmark
G18	Reinforcement learning in urban network traffic signal control: A systematic literature review	Noaeen	2022	Expert Systems with Applications	review (E3; used for chaining)
G19	Deep Reinforcement Learning for Intelligent Transportation Systems: A Survey	Haydari	2022	IEEE Transactions on Intelligent Transportation Systems	review (E3; used for chaining)
G20	Traffic Signal Control via Reinforcement Learning: A Review on Applications and Innovations	Michailidis	2025	Infrastructures	review (E3; used for chaining)

Appendix B — Extraction codebook

Source file: `data/protocol/codebook.csv`

table	field	section	supports	type	allowed_values	description
technical	paper_id	A	all	text		Stable paper ID (P001…)
technical	authors	A	RQ1	text		Authors
technical	year	A	RQ1	int		Publication year
technical	title	A	-	text		Title
technical	venue	A	RQ1	text		Journal or conference
technical	venue_type	A	RQ1	enum	journal; conference	Venue type
technical	doi	A	-	text		DOI
technical	country_first_author	A	RQ1	text		Country of first author's affiliation (pre-filled guess, verify)
technical	study_location	A	RQ1	text		City/country of the studied network, or 'synthetic'

table	field	section	supports	type	allowed_values	description
technical	application_context	B	RQ1	enum	isolated_intersection; arterial_corridor; grid_network; real_urban_network; other	Spatial setting of the controlled intersections
technical	n_intersections	B	RQ1	int		Largest number of RL-controlled intersections evaluated
technical	traffic_modes	B	RQ1;RQ6	multi	private_vehicles; pedestrians; cyclists; public_transport; emergency_vehicles; cav_connected_vehicles; freight; mixed	Road users explicitly modelled/prioritised
technical	objective_focus	B	RQ1	multi	efficiency; safety; emissions_energy; equity_fairness; priority; robustness	Objectives explicitly optimised
technical	rl_family	C	RQ2	enum	tabular; value_based_deep; policy_gradient; actor_critic; model_based; hierarchical; meta_rl; offline_rl; llm_rl_hybrid; other	RL algorithm family of the proposed method
technical	algorithm	C	RQ2	text		Named algorithm (e.g. DQN, D3QN, PPO, A2C, MAPPO)
technical	agent_architecture	C	RQ2	enum	single_agent; multi_agent_independent; multi_agent_cooperative; centralised_multi_intersection	Agent architecture
technical	coordination	C	RQ2	enum	none; neighbour_observation; explicit_communication; graph_attention; pressure_based; hierarchical; other	Coordination mechanism between intersections
technical	state_repr	C	RQ2	multi	queue_length; waiting_time; vehicle_count; occupancy; speed; position_matrix_image; pressure; current_phase; phase_duration; cv_trajectory; raw_camera; other	State features
technical	action_type	C	RQ2	enum	phase_selection; phase_switch_keep; phase_duration; cycle_split; offset; other	Action definition
technical	reward_type	C	RQ2	multi	queue_length; waiting_time; delay; pressure; throughput; travel_time; speed; emissions; safety; composite; other	Reward components
technical	function_approximator	C	RQ2	enum	table; mlp; cnn; rnn_lstm; gnn; attention_transformer; llm; other	Policy/value function approximator
technical	simulator	C	RQ3	multi	sumo; cityflow; vissim; aimsun; paramics; cbengine; custom; other; none	Simulation environment(s)
technical	sensing_source	C	RQ5	multi	loop_detector; camera; radar_lidar; connected_vehicle; simulator_ground_truth; other	Sensing assumed or used for the state
technical	network_source	F	RQ3	enum	synthetic; real	Road network geometry
technical	demand_source	F	RQ3	enum	synthetic; real_dataset; calibrated; field_live	Traffic demand used in evaluation

table	field	section	supports	type	allowed_values	description
technical	dataset_ids	F	RQ3;Step27	text		Benchmark datasets/scenarios, ';'-separated canonical names (e.g. jinan_3x4; hangzhou_4x4; ny_28x7; monaco; ingolstadt7; cologne8)
technical	baseline_classes	F	RQ3;RQ4	multi	fixed_time; actuated; max_pressure; sotl; scats_scoot; other_adaptive; rl; other	Classes of baselines compared
technical	vrl	F	RQ3;RQ4	vrl	0; 1; 2; 3; 4; 5	Validation Realism Level 0-5 (highest reached)
technical	transfer_test	F	RQ3	yn	Y; N	Evaluated on unseen scenarios / other simulator
technical	robustness_test	F	RQ3;RQ5	yn	Y; N	Detector noise/failure, demand surge or incident tests
technical	safety_constraints	F	RQ5	yn	Y; N	Min green, clearance, action masking or other constraints enforced
technical	n_seeds	F	QA5	text		Number of independent runs/seeds ('NR' if not reported)
technical	code_available	F	QA3	yn	Y; N	Code publicly available
technical	key_findings	G	RQ4	text		Main findings (1-2 sentences)
technical	advantages	G	RQ5	text		Advantages claimed
technical	limitations_authors	G	RQ5	text		Limitations stated by the authors
technical	barriers	H	RQ5	multi	sensing_data; sim_to_real_gap; safety_compliance; robustness; computation; communication; interpretability; scalability; multimodality; training_cost_sample_efficiency; regulation_institutional; none_reported	Deployment barriers discussed or evident
technical	mitigations	H	RQ5	text		Mitigations proposed for the barriers
technical	reviewer_strength	H	RQ6	text		Main methodological strength (reviewer)
technical	reviewer_weakness	H	RQ6	text		Main weakness (reviewer)
technical	research_gap_note	H	RQ6	text		Gap this study reveals or leaves open
technical	study_family_id	DEP	Step27	text		Same experiments/project/dataset family -> same ID (e.g. F01). Default = paper_id
technical	campaign_id	DEP	Step27	text		Field campaign or shared experiment ID, if any
technical	extracted_by	META	-	enum	R1; R2	Extractor
technical	verified_by	META	-	enum	R1; R2;	Verifier (20 % sample)
quantitative	entry_id	E	-	text		Q0001...
quantitative	paper_id	E	-	text		Paper ID
quantitative	scenario_id	E	Step27	text		Dataset/scenario canonical name for this number

table	field	section	supports	type	allowed_values	description
quantitative	metric	E	RQ4	enum	travel_time; waiting_time; delay; queue_length; stops; fuel; co2; emissions_other; safety_conflicts; throughput; speed; other	Outcome metric
quantitative	rl_method_name	E	RQ4	text		Name of the proposed RL method
quantitative	rl_value	E	RQ4	float		Value for the proposed RL method
quantitative	baseline_name	E	RQ4	text		Baseline name as in the paper
quantitative	baseline_class	E	RQ4	enum	fixed_time; actuated; max_pressure; sotl; scats_scoot; other_adaptive; rl; other	Baseline class
quantitative	baseline_value	E	RQ4	float		Value for the baseline
quantitative	reported_improvement_pct	E	RQ4	float		Use only if the paper gives the % improvement but not both raw values
quantitative	unit	E	RQ4	text		Unit (s, veh, veh/h, m, g…)
quantitative	rl_sd	E	Step26	float		SD/dispersion of RL value, if reported
quantitative	baseline_sd	E	Step26	float		SD of baseline value, if reported
quantitative	n_runs	E	Step26	int		Number of runs behind the values
quantitative	vrl	E	RQ4	vrl	0; 1; 2; 3; 4; 5	Validation level of this experiment (defaults to paper VRL)
quantitative	context	E	-	text		Demand level, peak/off-peak, test condition…
quantitative	source_location	E	audit	text		Page/table/figure, e.g. ‘Table 4, p. 9’
quantitative	extracted_by	META	-	enum	R1; R2	Extractor
quantitative	verified_by	META	-	enum	R1; R2;	Verifier (100 % of entries)

Appendix C — Methodological families (conceptual comparison)

Source file: `data/protocol/method_families.csv`

rl_family	principle	advantages	limitations	typical_validation	best_application
tabular	Look-up table of state-action values (Q-learning, SARSA)	Simple, interpretable, low compute, converges for small problems	Curse of dimensionality; needs coarse state discretisation; poor generalisation	Small synthetic simulations (VRL0-1)	Isolated intersections, embedded controllers with few states
value_based_deep	Neural network approximates Q-values (DQN, DDQN, D3QN, dueling, PER)	Handles rich states (queues, positions, images); discrete phase actions fit naturally; mature tooling	Overestimation, instability, sample inefficiency; exploration on live streets unsafe	SUMO/CityFlow real-data simulations (VRL1-2)	Phase selection at single or loosely coupled intersections
policy_gradient	Directly optimises a stochastic policy (REINFORCE, PPO, TRPO)	Stable updates (PPO); continuous or duration actions possible	High variance, many samples needed	Simulation (VRL0-2)	Phase-duration / cycle-split control

rl_family	principle	advantages	limitations	typical_validation	best_application
actor_critic	Actor (policy) + critic (value) learned jointly (A2C, A3C, DDPG, SAC, MA2C, MAPPO)	Lower variance than pure PG; natural for multi-agent CTDE	Hyper-parameter sensitivity; coordination overhead	Grid and real-network simulations (VRL1-2)	Network-level coordinated control
model_based	Learns or uses a traffic model for planning/imagination	Sample efficiency; can embed traffic-flow knowledge; safer exploration	Model bias; extra modelling effort	Simulation, occasionally HIL (VRL2-3)	Settings where interaction data are scarce
hierarchical	High-level policy sets goals/regions, low-level agents control signals	Scalability to large networks; temporal abstraction	Design complexity; credit assignment across levels	Large-network simulations (VRL1-2)	City-scale networks
meta_rl	Learns to adapt quickly to new intersections/demand (MetaLight, transfer)	Fast adaptation, generalisation across scenarios	Training cost; task distribution must cover reality	Cross-scenario simulation tests	Rolling out to many heterogeneous intersections
offline_rl	Learns from logged data without online exploration	No risky online exploration; uses historical detector logs	Distribution shift; requires good logged data	Logged real data + simulation (VRL2-3)	Bridging to field deployment
llm_rl_hybrid	Large language model reasons about traffic state; policy refined with RL/critic feedback	Generalisation, natural-language explanations, zero-shot transfer	Latency and compute; hallucination; limited safety guarantees	Simulation on real datasets (VRL2)	Decision support and interpretable control
other	Other or combined approaches				

Appendix D — Search log (PRISMA-S)

Source file: `data/protocol/search_log.csv`

search_id	wave	database	fields	filters	records_retrieved	run_by
S1	1	Scopus	TITLE-ABS-KEY	PUBYEAR 2019-2026 (in query); DOCTYPE ar, cp; LANGUAGE English		R1
S2	1	Web of Science Core Collection	TS	Editions SCI-EXPANDED, SSCI, ESCI, CPCI-S; DT Article, Proceedings Paper, Early Access; LA English		R2
S3	1	IEEE Xplore	All Metadata	Year 2019-2026; Conferences, Journals, Early Access Articles		R1